

# Daffodil International University

## PROJECT REPORT

Artificial Intelligence Lab

CSE-412

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Rock-Paper-Scissors

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Date of Submission: August 7, 2025

# Rock-Paper-Scissors

## 1. Introduction

This report details the creation of an adaptive AI for Rock-Paper-Scissors, a project for the Artificial Intelligence Lab course. The primary goal was to build an AI opponent that learns from a player's behavior to predict and counter their moves. The project demonstrates a clear progression in AI strategy through different learning models, implemented as varying difficulty levels. This document outlines the project's scope, algorithms, and individual contributions.

## 2. Project Description

We developed a web-based Rock-Paper-Scissors game where a player competes against an AI that analyzes their move history to improve its winning chances. The game features three difficulty levels—Easy, Medium, and Hard—each using a progressively more sophisticated prediction algorithm. The intuitive GUI provides real-time feedback, and a match is won when either player reaches five points.

## 3. Algorithm

The AI's strategy is to predict the player's next move and choose the counter-move. The prediction logic becomes more complex with each difficulty level.

### 3.1 Easy Difficulty: Frequency Analysis

- **Algorithm:** The AI counts the player's usage of rock, paper, and scissors, and predicts the player will use their most frequent move.
- **Justification:** This baseline algorithm assumes a simple player bias. It's easy to implement but also easy for a player to exploit.

### 3.2 Medium Difficulty: First-Order Markov Model

- **Algorithm:** The AI examines the player's last move and predicts they will make the same choice they most often made after that specific move in the past.

- **Justification:** This model identifies simple sequential patterns in human players (e.g., "if I won with rock, I'll play paper next"). It is more effective than frequency analysis but requires a history of moves.

### 3.3 Hard Difficulty: Second-Order Markov Model

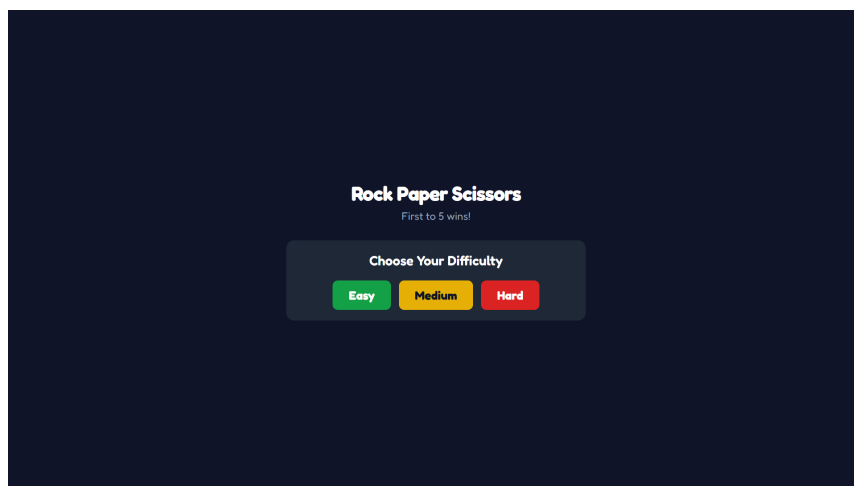
- **Algorithm:** The AI analyzes the player's last two moves to predict the next one, tracking move frequencies that follow specific two-move sequences. If history is insufficient, it defaults to the Medium AI's logic.
- **Justification:** This advanced model counters more complex player strategies and combos by looking at a longer sequence of moves, making the AI a very challenging opponent.

## 4. User Interaction

The application is GUI-based.

- **Input:** User clicks on buttons for difficulty ('Easy', 'Medium', 'Hard'), moves ('👊', '👐', '✂️'), or post-game actions ('Play Again').
- **Output:** The UI displays both choices, the round's result, and the updated score. The winner's choice is highlighted. A final message appears when a player reaches five points.

## 5. User Interface (UI)



*Figure 1: Difficulty Selection Screen*

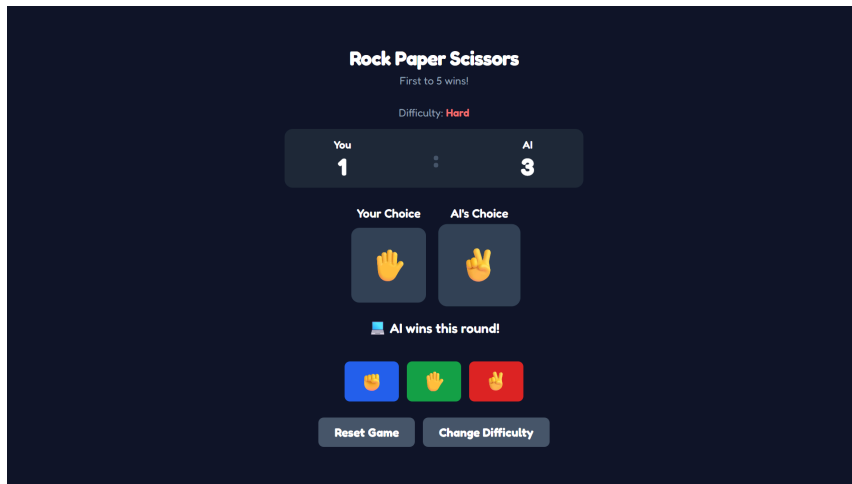


Figure 2: Main Game Interface

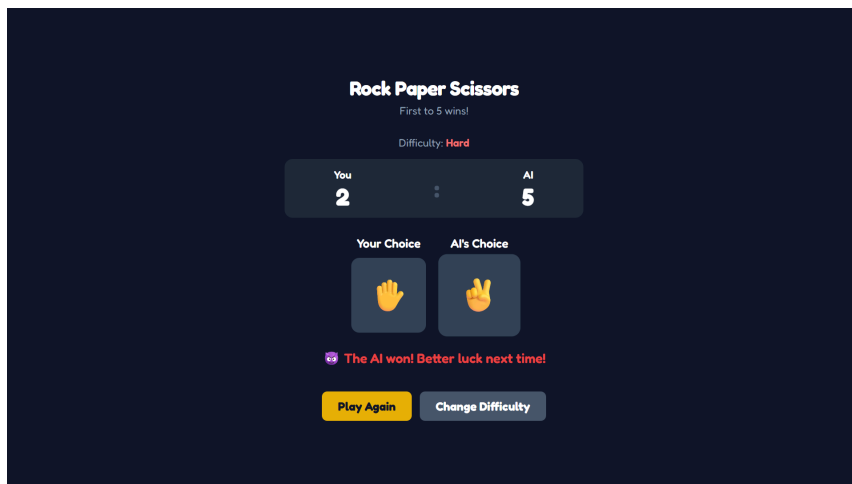


Figure 3: End Game Screen

## 6. Team Contributions

This project was a collaborative effort.

- **Raisul Kayes Raka:** Led the development of the AI prediction algorithms for all three difficulties, implemented the core game logic (round execution, scoring), conducted extensive debugging, and provided the final touches for an improved UI/UX.
- **Md. Saiful Hasan:** Handled all frontend development, designing the UI with HTML/CSS and writing the JavaScript for DOM manipulation and game state management.

Both members contributed to initial planning and final integration.

## 7. Conclusion

This project successfully created an engaging Rock-Paper-Scissors game with an intelligent AI. The tiered difficulty levels effectively demonstrate machine learning principles, from simple analysis to complex Markov models, with a noticeable increase in challenge.

Future work could involve implementing a mixed strategy for the AI to avoid predictability, using a more advanced model like a neural network, or adding persistent memory to allow the AI to learn a player's style across multiple sessions.